

SAFETY DATA SHEET

according to the Globally Harmonized System and Korean regulation

Dissolvine Na2

Version 2 Revision Date: 20.03.2022 Print Date: 23.06.2023 KR / EN

1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

Product Information

Trade name : Dissolvine Na2

Use of the Substance/Mixture : Specific use(s): Industrial and Cosmetic

Company : Nouryon Surface Chemistry AB
Stenunge Alle 3
SE 444 85 Stenungsund
Sweden

Telephone : +4630385000
Telefax : +4630384659
E-mail address : Regulatory.Affairs@nouryon.com
Emergency telephone number : 020 99 60 00 Kemiakuten, SE +31 57 06 79 211 24 hours emergency response number

2. HAZARDS IDENTIFICATION

GHS Classification

Acute toxicity, Category 4, Inhalation
Specific target organ toxicity - repeated exposure, Category 2, Inhalation, Respiratory Tract

GHS label elements

Hazard pictograms : 

Signal word : Warning

Hazard statements : H332 Harmful if inhaled.
H373 May cause damage to organs (Respiratory Tract) through prolonged or repeated exposure if inhaled.

Precautionary statements : **Prevention:**
P260 Do not breathe dust or fume.
P271 Use only outdoors or in a well-ventilated area.
Response:

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P304 + P340 + P312 IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Call a POISON CENTER or doctor/ physician if you feel unwell.
P314 Get medical advice/ attention if you feel unwell.

Disposal:

P501 Dispose of contents and container according to wastes control act.

Other hazards which do not result in classification

Risk of dust explosion.

No further data available.

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3. COMPOSITION/INFORMATION ON INGREDIENTS

Common Name : Ethylenediaminetetraacetic acid, disodium salt, dihydrate
Chemical nature : Substance

Hazardous substance

Chemical name	CAS-No.	Classification	Concentration [%]	KECI Number
Ethylenediaminetetraacetic acid, disodium salt Ethylenediaminetetraacetic acid, disodium salt	6381-92-6	Acute Tox. 4; H332 STOT RE 2; H373	100	KE-13651

For the full text of the H-Statements mentioned in this Section, see Section 16.

4. FIRST AID MEASURES

General advice : Move out of dangerous area.
Consult a physician.
Show this safety data sheet to the doctor in attendance.

Inhalation : Rinse nose and mouth with water.

Skin contact : Take off contaminated clothing and shoes immediately.

Eye contact : Rinse with plenty of water.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.

Ingestion : Clean mouth with water and drink afterwards plenty of water.
Never give anything by mouth to an unconscious person.
Obtain medical attention.

Notes to physician

Symptoms : The symptoms and effects are as expected from the hazards as shown in section 2. No specific product related symptoms are known.

Risks : Harmful if inhaled.
May cause damage to organs through prolonged or repeated exposure if inhaled.

Treatment : Treat symptomatically.

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5. FIREFIGHTING MEASURES

- | | | |
|---|---|--|
| Suitable extinguishing media | : | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| Specific hazards during firefighting / Specific hazards arising from the chemical | : | Do not allow run-off from fire fighting to enter drains or water courses.
Risks of ignition followed by flame propagation or secondary explosions shall be prevented by avoiding accumulation of dust, e.g. on floors and ledges. |
| Combustion products | : | Nitrogen oxides (NO _x) |
| Special protective equipment for firefighters | : | In the event of fire, wear self-contained breathing apparatus. |
| Further information | : | Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. |

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

- | | | |
|--|---|---|
| Personal precautions | : | Use personal protective equipment.
Wear respiratory protection.
Avoid dust formation.
Avoid breathing dust.
Ensure adequate ventilation. |
| Emergency measures on accidental release | : | Evacuate personnel to safe areas.
Only qualified personnel equipped with suitable protective equipment may intervene.
Prevent unauthorised persons entering the zone. |
| Environmental precautions | : | Do not flush into surface water or sanitary sewer system.
If the product contaminates rivers and lakes or drains inform respective authorities. |
| Methods for cleaning up /
Methods for containment | : | Pick up and arrange disposal without creating dust.
Sweep up and shovel.
Keep in suitable, closed containers for disposal. |
| Reference to other sections | : | For disposal considerations see section 13.

For personal protection see section 8. |

7. HANDLING AND STORAGE

Handling

- | | | |
|-------------------------|---|---|
| Advice on safe handling | : | For personal protection see section 8.
Avoid formation of respirable particles.
Do not breathe vapours/dust.
Keep away from heat/sparks/open flames/hot surfaces. No |
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smoking.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
Dispose of rinse water in accordance with local and national regulations.

Advice on protection against fire and explosion : Provide appropriate exhaust ventilation at places where dust is formed.
No sparking tools should be used.

Storage

Requirements for storage areas and containers : Prevent unauthorized access.
Keep in a dry place.
Store at room temperature in the original container.
Keep container tightly closed.

Other data : No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value	Control parameters	Update	Basis	Form of exposure
Dust		TWA	4 mg/m ³		KR OEL	Total dust

ACGIH: American Conference of Governmental Industrial Hygienists
BEI: Biological Exposure Index
MAC: Maximum Allowable Concentration
NIOSH: National Institute for Occupational Safety and Health
OEL: OEL: Occupational exposure limit.
STEL: Short term exposure limit
TWA: Time Weighted Average

Appropriate engineering controls

Provide appropriate exhaust ventilation at places where dust is formed.

Personal protective equipment

Respiratory protection : Half mask with a particle filter P2 (EN 143)

Eye/face protection : Tightly fitting safety goggles

Skin and body protection : Protective suit

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice.
When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.

Environmental exposure controls

General advice : Do not flush into surface water or sanitary sewer system.
If the product contaminates rivers and lakes or drains inform respective authorities.

9. PHYSICAL AND CHEMICAL PROPERTIES**Appearance**

Form : powder
Colour : white
Odour : odourless
Odour Threshold : Not applicable

Safety data

pH : 4 - 5 1% (water)
Melting point : Decomposes before melting.
Boiling point/boiling range : Not applicable
Flash point : Not applicable
Ignition temperature : ≥ 200 °C
Method: Auto-ignition of a 5mm dust layer according to EN 50281-2-1
Evaporation rate : Not applicable
Flammability (solid, gas) : Not classified as a flammability hazard
May form combustible dust concentrations in air during processing, handling or other means.
Flammability (liquids) : Not applicable
Lower explosion limit : ≥ 40 g/m³
Upper explosion limit : Not applicable
Vapour pressure : Not applicable
Relative vapour density : Not applicable
Relative density : No data available
Bulk density : ca. 700 kg/m³
Water solubility : ca. 100 g/l
Solubility in other solvents : No data available

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Partition coefficient: n-octanol/water	: log Pow: < 0
Auto-ignition temperature	: No data available
Decomposition temperature	: No data available
Viscosity, dynamic	: Not applicable
Viscosity, kinematic	: Not applicable
Explosive properties	: Not explosive
Oxidizing properties	: Not classified as oxidising.

This safety datasheet only contains information relating to safety and does not replace any product information or product specification.

10. STABILITY AND REACTIVITY

Conditions to avoid	: None known.
Materials to avoid	: None known.
Hazardous decomposition products	: Carbon oxides nitrogen oxides (NO _x)
Thermal decomposition	: No data available
Reactivity	: Stable under normal conditions.
Chemical stability	: Stable under recommended storage conditions.
Hazardous reactions	: Dust may form explosive mixture in air.

11. TOXICOLOGICAL INFORMATION

PRODUCT INFORMATION:

Hazard Summary

Acute toxicity	: Harmful if inhaled.
Skin corrosion/irritation	: Not classified based on available information.
Serious eye damage/eye irritation	: Not classified based on available information.
Respiratory or skin sensitisation	: Respiratory sensitisation: Not classified based on available information. Skin sensitisation: Not classified based on available information.
Germ cell mutagenicity	: Not classified based on available information.
Carcinogenicity	: Not classified based on available information.

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Reproductive toxicity	: Not classified based on available information.
STOT - single exposure	: Not classified based on available information.
STOT - repeated exposure	: May cause damage to organs (Respiratory Tract) through prolonged or repeated exposure if inhaled.
Aspiration hazard	: Not classified based on available information.

Potential Health Effects

Inhalation	: Product dust may be irritating to the respiratory system. Harmful if inhaled.
Skin	: Product dust may be irritating to skin.
Eyes	: Product dust may be irritating to eyes.
Ingestion	: May be harmful if swallowed.
Aggravated Medical Condition	: None known.
Symptoms of Overexposure	: The symptoms and effects are as expected from the hazards as shown in section 2. No specific product related symptoms are known.

Toxicology Assessment

Further information	: No further data available.
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Test result

Acute inhalation toxicity	: Acute toxicity estimate : 1.68 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: Calculation method
Target Organ Systemic Toxicant - Repeated exposure	: Exposure routes: Inhalation Target Organs: Respiratory Tract The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 2.

TOXICOLOGY DATA FOR THE COMPONENTS:

Toxicology Assessment

Component: Nitrilotriacetic acid, disodium salt

CMR effects	: Carcinogenicity: Limited evidence of a carcinogenic effect.
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Test result

Component: Ethylenediaminetetraacetic acid, disodium salt

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Acute oral toxicity	: LD50: > 2,000 mg/kg Species: Rat
Acute inhalation toxicity	: LC50 (Rat): > 1 - 5 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 412
Skin irritation	: Species: Rabbit Result: No skin irritation Method: OECD Test Guideline 404
Eye irritation	: Species: Rabbit Result: No eye irritation Method: OECD Test Guideline 405
Sensitisation	: Maximisation Test Species: Guinea pig Result: Does not cause skin sensitisation. Method: OECD Test Guideline 406
Germ cell mutagenicity Genotoxicity in vitro	: Result: negative Method: Mutagenicity (Salmonella typhimurium - reverse mutation assay)
Genotoxicity in vivo	: Chromosome aberration test in vivo Species: Mouse Method: OECD Test Guideline 474 Result: negative
Carcinogenicity	: Species: Rat Application Route: Ingestion Result: Not classified due to data which are conclusive although insufficient for classification. Read-across (Analogy)
Reproductive toxicity	: Species: Rat NOAEL: F1: > 250 mg/kg, Read-across (Analogy), Information taken from reference works and the literature.
Target Organ Systemic Toxicant - Single exposure	: Based on available data, the classification criteria are not met.
Target Organ Systemic Toxicant - Repeated exposure	: Exposure routes: Inhalation Target Organs: Respiratory Tract The substance or mixture is classified as specific target organ toxicant, repeated exposure, category 2.
Aspiration toxicity	: Not classified due to data which are conclusive although insufficient for classification.

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Component: Nitrilotriacetic acid, disodium salt

Acute oral toxicity : LD50: 1,000 - 2,000 mg/kg
Species: Rat
Method: OECD Test Guideline 401

Eye irritation : Result: Irritating to eyes.

12. ECOLOGICAL INFORMATION**PRODUCT INFORMATION:****Ecotoxicology Assessment**

Additional ecological information : None known.

Test result**Elimination information (persistence and degradability)**

Bioaccumulation : Not expected considering the low log Pow value.

Mobility : Adsorption to the solid soil particles is not expected.

Biodegradability : Not readily biodegradable, but will degrade after a longer period.

Further information on ecology

Biochemical Oxygen Demand (BOD) : No data available

COMPONENTS:**Test result****Component: Ethylenediaminetetraacetic acid, disodium salt****Ecotoxicity effects**

Toxicity to fish : LC50: > 100 mg/l
Exposure time: 96 h
Species: Fish
Read-across (Analogy)

Toxicity to daphnia and other aquatic invertebrates : EC50 (calculated): 140 mg/l
Exposure time: 48 h
Species: Daphnia magna (Water flea)
Method: DIN 38412

Toxicity to bacteria : EC50: > 500 mg/l
Exposure time: 30 min
Species: activated sludge
Test Type: Respiration inhibition
Method: OECD Test Guideline 209

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Toxicity to fish (Chronic toxicity) : NOEC: > 25.7 mg/l
Exposure time: 35 d
Species: Danio rerio (zebra fish)
Test Type: flow-through test
Method: OECD Test Guideline 210
Read-across (Analogy)

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEC: > 25 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)

Elimination information (persistence and degradability)

Bioaccumulation : Not expected considering the low log Pow value.

Mobility : Adsorption to the solid soil particles is not expected.

Biodegradability : Not readily biodegradable, but will degrade after a longer period.

Further information on ecology

Biochemical Oxygen Demand (BOD) : No data available
Chemical Oxygen Demand (COD) : 630 mg/g

Component: Nitrilotriacetic acid, disodium salt

Ecotoxicity effects

Toxicity to fish : LC50: > 100 mg/l
Exposure time: 96 h
Species: Pimephales promelas (fathead minnow)
Read-across (Analogy)

Elimination information (persistence and degradability)

Bioaccumulation : Bioaccumulation is unlikely.

Mobility : Adsorption to the solid soil particles is not expected.

Biodegradability : Expected to be biodegradable

Further information on ecology

Biochemical Oxygen Demand (BOD) : No data available

13. DISPOSAL CONSIDERATIONS

Product : Do not dispose of waste into sewer.

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Do not contaminate ponds, waterways or ditches with chemical or used container.
Hazardous waste
Dispose of contents/container in accordance with local regulation.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.

14. TRANSPORT INFORMATION

International Regulations

ADR

Not regulated as a dangerous good

UNRTDG

Not regulated as a dangerous good

IATA-DGR

Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Further information for transport

Not classified as dangerous in the meaning of transport regulations.

National Regulations

KR_DG

Not regulated as a dangerous good

Refer to section 15 for specific national regulation.

15. REGULATORY INFORMATION

Notification status

DSL : YES. All components of this product are on the Canadian DSL
AICS : YES. On the inventory, or in compliance with the inventory
ENCS : YES. On the inventory, or in compliance with the inventory
ISHL : YES. On the inventory, or in compliance with the inventory
KECI : YES. On the inventory, or in compliance with the inventory
PICCS : YES. On the inventory, or in compliance with the inventory
IECSC : YES. On the inventory, or in compliance with the inventory
TCSI : YES. On the inventory, or in compliance with the inventory
TSCA : YES. All chemical substances in this product are either listed on the TSCA Inventory or in compliance with a TSCA Inventory exemption.

For explanation of abbreviation see section 16.

National regulatory information

Regulation under the Occupational Safety and Health Act

Harmful Substances Prohibited from Manufacturing

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Not applicable

Harmful Substances Required Permission for Manufacture

Not applicable

Harmful Agents to be kept below Occupational Exposure Limits

Not applicable

Hazardous substances requiring management

Not applicable

Special Management Materials

Not applicable

Controlled Substances Subject to Environment Monitoring

Not applicable

Controlled Substances Subject to Health Examination

Not applicable

Priority Existing Chemicals

Not applicable

Act on the Registration and Evaluation, etc. of Chemical Substances, Chemicals Control Act

Toxic Chemicals

Not applicable

Restricted Chemicals

Not applicable

Prohibited Chemicals

Not applicable

Toxic Release Inventory

Chemical name	CAS-No.	Group	Threshold limits [%]
None			

Accident Precaution Chemicals

Not applicable

Dangerous Substances Safety Management Act

Not Applicable to Dangerous Materials

Wastes Control Act

Industrial waste

Follow article 13 of the act to dispose the product waste

Further information : none

16. OTHER INFORMATION

Full text of H-Statements

H302 : Harmful if swallowed.
H319 : Causes serious eye irritation.
H332 : Harmful if inhaled.
H351 : Suspected of causing cancer.
H373 : May cause damage to organs through prolonged or repeated exposure if inhaled.

Full text of other abbreviations

KR OEL : Harmful Agents to be kept below Occupational Exposure

Limits

KR OEL / TWA : Time weighted average

AICS - Australian Inventory of Chemical Substances; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; CPR - Controlled Products Regulations; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

Further information

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.